Text Summarization System

1. Description of the System

The system is an end-to-end NLP pipeline designed to condense long textual documents into concise summaries while retaining key information. It employs a **Multi-Model Approach**, allowing users to compare three different summarization strategies:

- **Statistical (TF-IDF):** A baseline extractive method focusing on word frequency.
- **Semantic Hybrid:** An advanced extractive method combining statistical importance with deep learning embeddings.
- **Generative (BART):** A state-of-the-art abstractive method that rewrites the text to ensure linguistic fluency.

The system features a **Streamlit-based GUI** for real-time interaction and a comprehensive **Evaluation Dashboard** to measure performance using industry-standard metrics.

2. Feature Extraction

We implemented two main types of feature extraction to represent text data:

- **TF-IDF:** Used to calculate sentence importance based on word frequency across the document (Statistical features).
- **Word Embeddings:** Utilized `all-MiniLM-L6-v2` (Transformer-based) to capture deep semantic meaning and context of the sentences (Semantic features).

3. Baseline Method

The baseline consists of **Extractive Summarization using TF-IDF**. This method ranks sentences by calculating their importance based on the TF-IDF scores of the words they contain and then selecting the **top important sentences** to form the summary.

4. Advanced Method

Our advanced approach includes two sophisticated methods:

- **Hybrid Semantic Summarizer:** Combines both TF-IDF and Transformer embeddings to rank sentences, capturing both keyword importance and global context.

- **Transformer-based Summarization:** Deployed a pre-trained **BART model** (`facebook/bart-large-cnn`) for **Abstractive Summarization**. Unlike extractive methods, this model generates new, human-like sentences.

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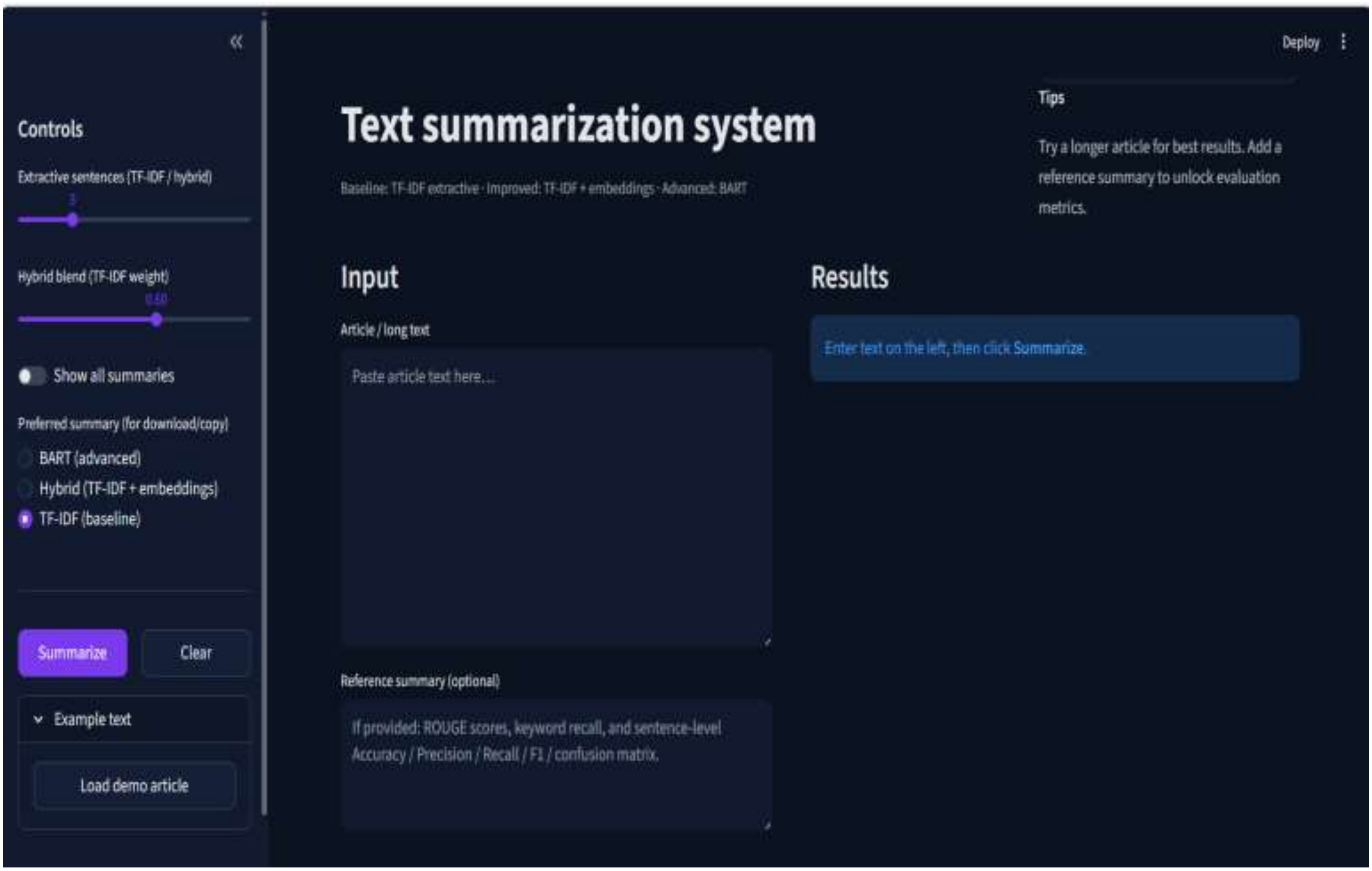
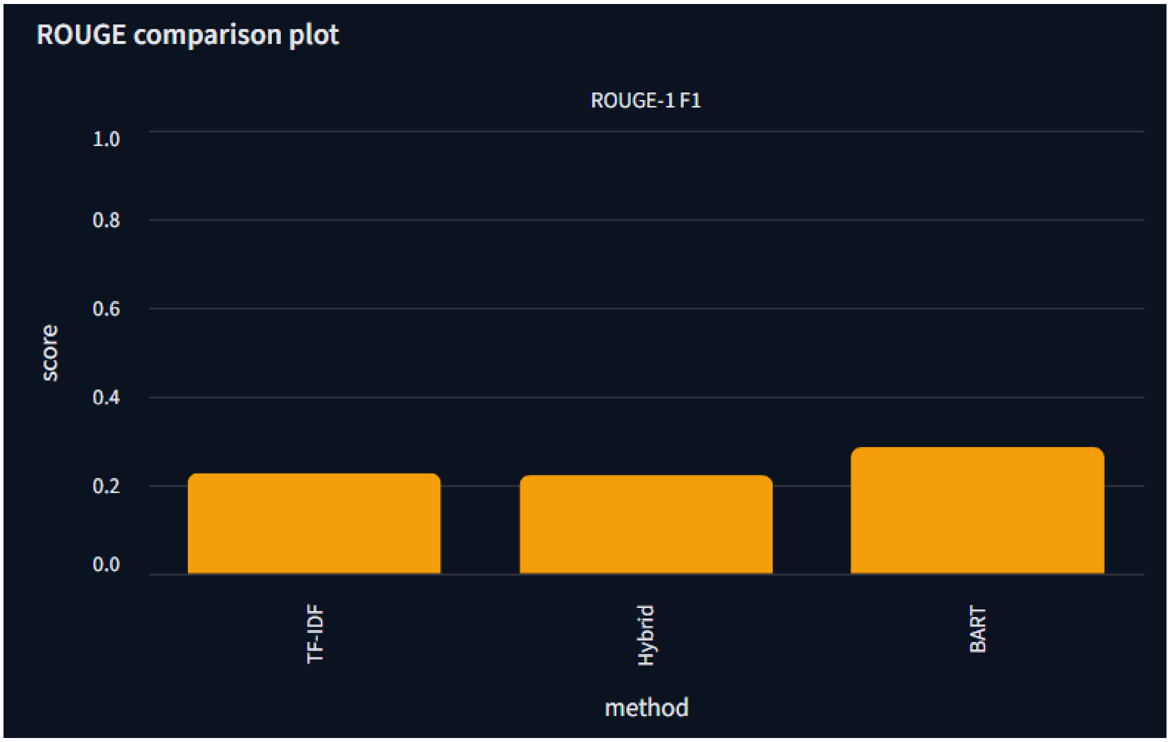
5. Results and Visualizations

A. System Interface Overview



B. Performance Metrics (ROUGE Scores)



Overlap metrics vs reference

ROUGE-L (TF-IDF)	ROUGE-L (hybrid)	ROUGE-L (BART)
0.1136	0.1778	0.1758
ROUGE-1 F1: 0.2273	ROUGE-1 F1: 0.2222	ROUGE-1 F1: 0.2857

C. Efficiency (Compression Ratio & Keyword Recall)

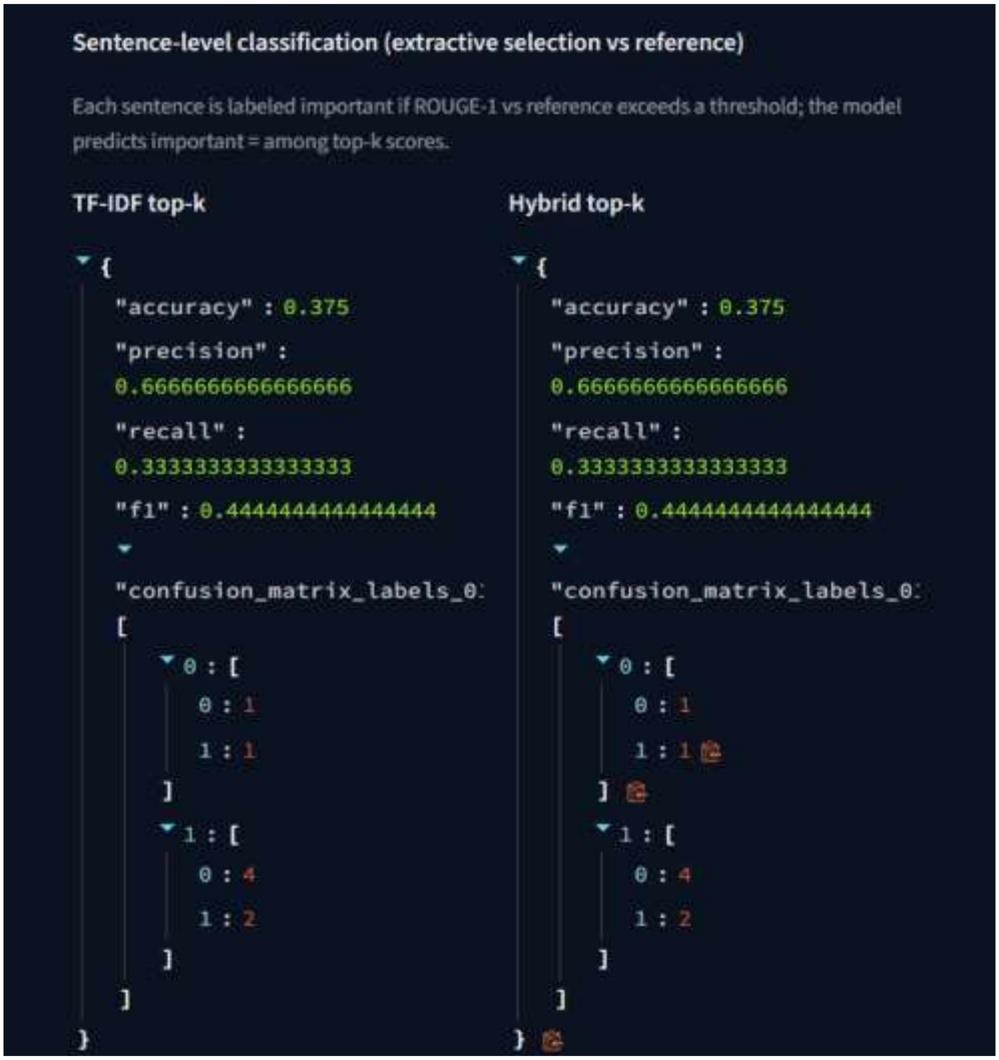
Compression ratio (summary words / article words)

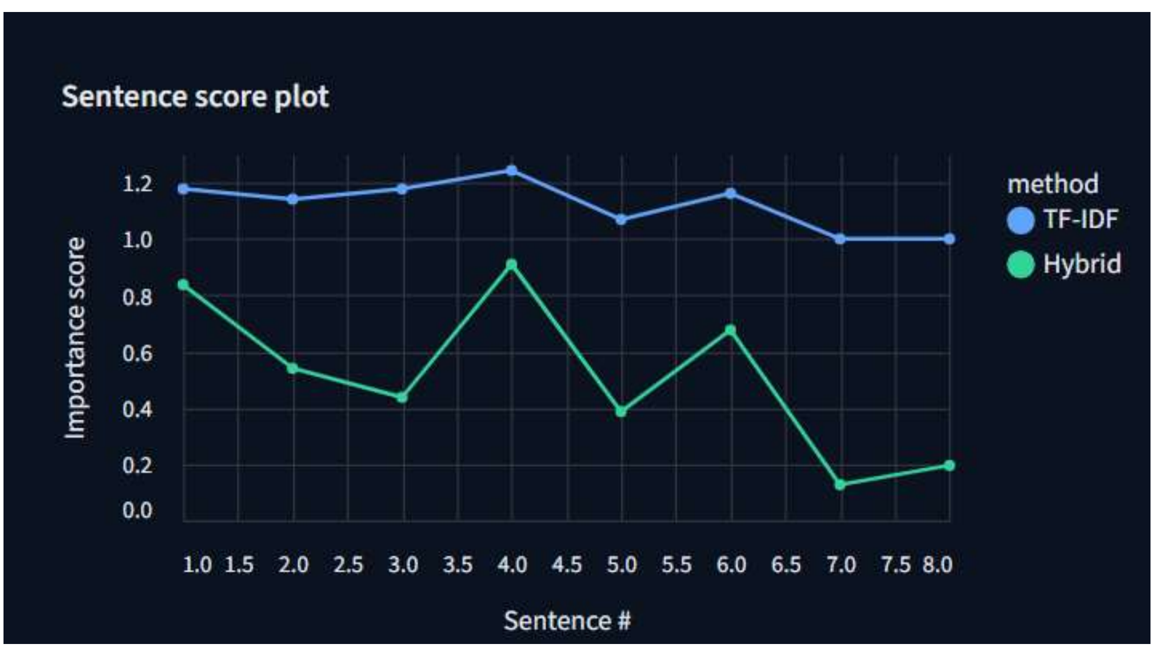
```
{
  "TF-IDF" : "0.348"
  "Hybrid" : "0.362"
  "BART" : "0.383"
}
```

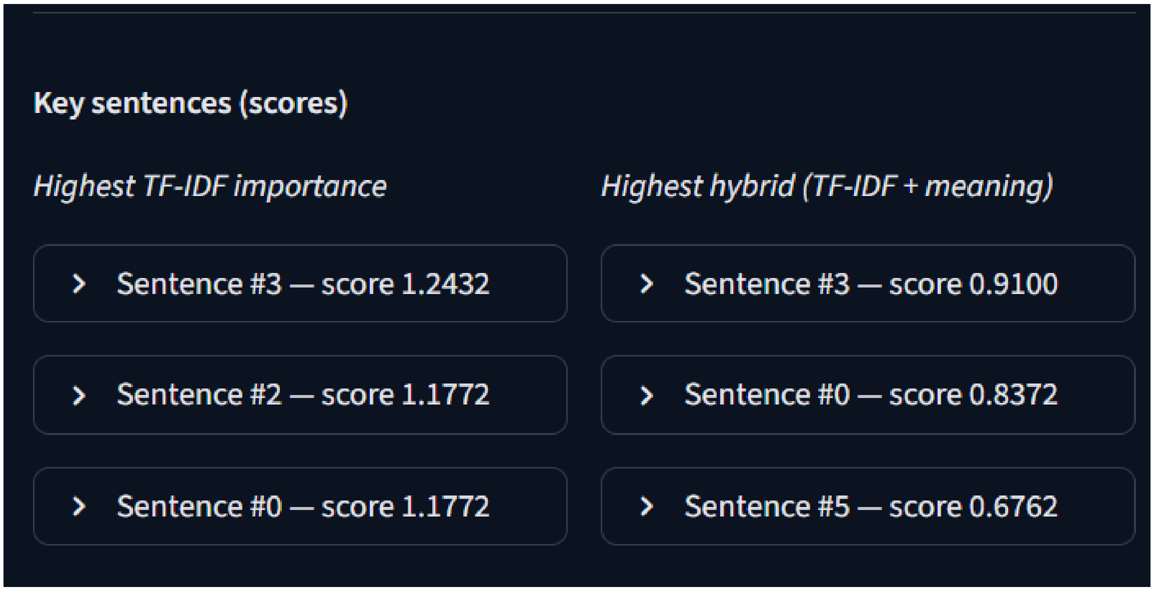
Keyword recall (overlap of top TF-IDF terms)

```
{
  "TF-IDF" : "0.533"
  "Hybrid" : "0.600"
  "BART" : "0.533"
}
```

D. Sentence Classification and Score Results







6. Conclusion

The project successfully demonstrates the power of combining statistical methods with Deep Learning. The **Hybrid Model** proved to be the most reliable for accuracy, while the **BART Model** excels in generating natural, fluent summaries.